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Is it all about the stars? Spending on star players and success in the National Football League

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Abstract

The National Football League (NFL) introduced a salary cap in 1994 which significantly affected the payroll efficiency of teams within the league. This paper uses a robust directional distance Data Envelopment Analysis (DEA) model to measure and evaluate the payroll efficiency of the 32 teams of the NFL from 2011-2020. The focus is on investigating if teams are more payroll efficient when spending on star players or spreading their expenditure more equally across their squad. Four DEA-models were defined to analyse spending payroll efficiency relating to the overall allocation of salary, star offensive players, star defensive players, and starting quarterbacks. Inputs were relevant salary cap information, while outputs were performance statistics unique to each category. A relationship was discovered between payroll efficiency in each category and qualification for the playoffs and post-season success. Efficient expenditure on star defensive players was especially important.

Key words: Data Envelopment Analysis; NFL; Star Players; Salary Cap; Performance Evaluation.

1. Introduction

The NFL is America's most popular sport and had seven of the world's top 15 most valuable sports teams in 2020, including the most valuable team, the Dallas Cowboys. One reason for the NFL's success has been its structure which lends itself to relative parity among the league's 32 teams. The annual NFL Draft allows teams with the worst records to select the most-promising prospects emerging from college football, while a salary cap was introduced in 1994 to restrict how much teams can spend on their squad. This cap number has steadily risen from $38 million at its inception to $198.2 million in 2020, allowing players to also benefit from the ever-growing, lucrative television contracts. However, teams are still forced to manage their salary cap closely and choose the optimal areas of the team in which to invest.

Salary cap percentages reveal the value that each NFL team places on different positions, allowing for comparisons with other teams (Moore, 2018). When a team invests heavily in a particular position, they must make up for it by finding value elsewhere. According to Moore (2018, pg. 70), in terms of positional importance, "the total quality of an NFL team is four parts offense, three parts defense, and one part special teams". The most crucial and most costly position is the quarterback, making players who enhance the quarterback's performance on offense or hinder it on defense highly valuable assets. These key positions are typically quarterback, left tackle, outside wide receiver, cornerback and pass rusher.

Previously, Data Envelopment Analysis (DEA) models have been used to analyse overall payroll efficiency in the NFL, as well as sub-categories including offense, defense and the impact of injuries on highly paid players. However, no research has been conducted regarding spending on star players. Therefore, this study aims to evaluate which NFL teams have maximised their payroll efficiency in relation to star players over a specific time period. For instance, the New England Patriots won six Super Bowls from 2001 to 2018, yet they rarely paid their star quarterback, Tom Brady, near the top of the market. For example, for their Super Bowl winning season in 2016, Brady was paid only 8.87% of the cap (Moore, 2018), ranking his salary 18th in the league. This paper will determine if this strategy results in improved payroll efficiency.

In this research, a robust order-m directional distance DEA-model is defined with multiple inputs and outputs to evaluate the payroll efficiency of the 32 franchises of the NFL in relation to salary cap management. Specifically, this research aims to determine if teams are more payroll efficient when allocating a significant proportion of resources on a small elite band of players or to a balanced roster. Four models were established to analyse payroll efficiency in relation to: (i) overall salary allocation; (ii) star offensive players; (iii) star defensive players; and (iv) starting quarterbacks. Inputs for the models consisted of salary expenditure figures pertaining to the specific categories, while outputs included regular season victories and team performance statistics such as Points Per Game (PPG) and Points Allowed Per Game (PAG). The directional distance version of the DEA-model was selected to consider the mix of desirable and undesirable performance factors in assessing the payroll efficiency of each NFL team in effectively managing the salary cap. Data from 10 NFL seasons (i.e., the period 2011-2020) were analysed. The results were used to explore the relationship between payroll efficiency and success in reaching the playoffs and winning titles in the NFL.

The remainder of this paper is organised as follows. Section 2 explores the literature on NFL, the NFL salary cap and the use of the DEA method to measure efficiency in American football

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and other popular sports. Section 3 presents the team salary information, offensive and defensive performance statistics that are used to measure NFL team payroll efficiency. Section 4 briefly discusses the basic DEA-method and the robust directional distance variant of the DEA-method used in this paper to measure and evaluate team payroll efficiency under salary cap restrictions. Section 5 presents, discusses and analyses the results. Finally, section 6 concludes and provides some suggestions for further research on this particular topic.

2. Literature

2.1. NFL salary cap

Between the inception of the Super Bowl in January 1967 and the creation of the salary cap in 1994, certain popular teams tended to succeed in the NFL. The Green Bay Packers (four titles), Pittsburgh Steelers (four titles), San Francisco 49ers (four titles) and Dallas Cowboys (three titles) dominated the NFL in the 1960s, 70s, 80s and 90s respectively. These "dynasties" were able to dominate in an era before the implementation of a hard salary cap and the introduction of free agency, which allowed players to move freely between teams. To increase parity, a salary cap of $38 million was introduced in 1994, changing the face of the NFL forever (McGee, Van Scyoc & Burnett, 2012). The NFL has experienced huge increases in television revenue since the 1990s, leading the salary cap to sky rocket to $198.2 million in 2020.

The main outcome of the salary cap was increased parity among the 32 teams, where the financial structure does not favour specific franchises and teams cannot overspend to achieve goals (Palafox-Alcantar & Vargas-Hernandez, 2015). Specifically, the creation of the salary cap prevented franchises with large profits from recruiting a large number of star players. Extensive revenue sharing in the NFL led to the depression of player salaries, while the imposition of a salary cap allows teams to act as a cartel, leading to the increased exploitation of players (Vrooman, 2000). The NFL is not a completely free market, however (Roach, 2018). A price floor exists for minimum salaries based on a player's experience, while the salary cap serves as a payroll ceiling for overall spending. Certain bonus payments (e.g. signing bonus) can be spread across the lifetime of the contract. Other bonuses can be labelled as likely or not likely to be earned. If earned, these would affect the player's cap figure. Players who are drafted in the early stages of the seven-round NFL draft are compensated at a higher level on their initial four-to-five-year contract than players drafted in the later rounds due to performance expectations. NFL contracts are not fully guaranteed like they are in baseball or basketball which gives NFL teams greater flexibility. If a player's productivity no longer meets his salary, teams can jettison the player without hurting their financial position. The performance of an individual in a team sport can be difficult to ascertain (Roach, 2018). A new rookie pay scale was introduced with the 2011 Collective Bargaining Agreement. This meant unproven, first-round rookies entering the NFL were not guaranteed exorbitant salaries on their first contract. For example, the number one overall picks in 2010 (Sam Bradford) and 2011 (Cam Newton) signed $78 million and $22.025 million contracts, respectively.

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The introduction of the salary cap significantly impacted the payroll efficiency of NFL teams. The previous financial set-up that encouraged inefficient payrolls forced some teams to overspend on salaries, which in turn hurt other teams in the league (Palafox-Alcantar & Vargas-Hernandez, 2015). The salary cap, by its nature, drives payroll efficiency as teams have to be more careful about how they spend their money. Research has shown that 60% of observable variance in player salary is related to skill, ability and on-field performance (Frick, 2011), while the most expensive players tend to have the largest impact on the game (Szymanski & Wilkinson, 2016). Teams can vary significantly technically, tactically, physically and psychologically depending on the players (Zambom-Ferraresi, Iraizoz & Lera-Lopez, 2019). Therefore, it can be deduced that investing in an optimum facet of an NFL team is crucial for success.

Efficiency analysis in sports is becoming increasingly common and is critical for identifying market inefficiencies when managing valuable resources. Baseball teams with no salary cap tend to be less efficient in payroll management compared to NFL franchises. Berri, Brook and Schmidt (2007) identified a market inefficiency in basketball, where teams overvalue player points compared to other important statistics when determining salaries. In the NFL, teams significantly overvalued picks in the early rounds of the NFL draft due to psychological factors (Massey & Thaler, 2013). Meanwhile, players selected in the early rounds of the NFL draft tended to have more productive careers if they came from less famous schools (Hendricks, DeBrock & Koenker, 2003). Other possible reasons for payroll inefficiency of NFL teams include injuries to high-priced players and poor management skills, such as overspending on lacklustre talent (Einolf, 2004). Finally, the sunk-cost fallacy remains a significant determinant of how much game time is afforded to individual players (Keefer, 2015). All these factors highlight the importance of payroll efficiency analysis for NFL teams.

2.2. DEA research on NFL

The DEA method has been utilized in several interesting studies on the efficiency of NFL teams. Chen, Huang, Hung, Chen, Wu and Huang (2010) employed grey relational analysis to rank the fighting ability and DEA to assess the economic efficiency of the 32 teams in the NFL. The grey relational analysis confirmed that 11 of the 12 teams that made the playoffs in 2005 were among those with the highest fighting ability. The DEA-analysis showed that nine teams could have reduced spending and increased ticket sales to enhance economic efficiency. Nevertheless, teams with poor economic efficiency in terms of finances still managed to reach the playoffs, such as the Carolina Panthers, Seattle Seahawks, New England Patriots, Cincinnati Bengals, Pittsburgh Steelers, Indianapolis Colts and Denver Broncos. Grey analysis, cluster analysis and discriminant analysis were all effective tools for predicting potential Super Bowl champions.

Einolf (2004) used DEA to analyse payroll efficiency of all NFL teams from 1981 to 2000 and MLB teams from 1985-2001. Player salaries were used as inputs, while on-the-field performance statistics including number of wins were used as outputs. In the NFL model, salaries were split between offensive and defensive players, while the outputs included total wins and the widely-used team performance statistics Yards Per Offensive Play (YOP) and Yards Per Defensive Play (YDP). An NFL team was deemed payroll inefficient when another team produced more wins, higher YOP and lower YDP, while also paying lower salaries. Crucially, NFL teams tended to be more payroll efficient than their MLB counterparts.

Revenue sharing and the introduction of a salary cap to the NFL in 1994 were key reasons for the discrepancies. Payroll efficiency improved for NFL teams after the introduction of the salary cap, as high payrolls were heavily correlated with payroll inefficiency before the cap's introduction. For example, Al Davis of the Los Angeles and Oakland Raiders overspent heavily on players in search of a Super Bowl title, leading the Raiders to spend the highest amount on salaries for seven consecutive unsuccessful years, making Davis the league's most payroll inefficient owner. The Steelers, who had a more consistent and less aggressive philosophy, were the most payroll efficient team in the study. This correlation disappeared after the introduction of the salary cap in 1994 (Einolf, 2004).

Conversely, high-spending teams from big markets in Major League Baseball (MLB) were found to be payroll inefficient (Einolf, 2004). Essentially, big-market teams in the MLB could overspend on star players which led to MLB championships. From 1995 to 2001, teams from the lower half of the leagues in terms of total salary only won five playoff games, while teams from the top 25% highest-spending teams won six out of seven World Series titles. After the salary cap was introduced in the NFL, a higher percentage of teams with low-salary output made the playoffs and won Super Bowls. Teams may have different metrics for success, however, so multiple payroll inefficient seasons followed by a Super Bowl victory may be deemed successful by certain teams but would be payroll inefficient in this study. Overall, payroll inefficiency was mainly caused by owner mismanagement and the financial system did not favour specific teams as no team could overspend to ensure victory. These points will be key foundational points for this research.

The NFL has players who specialise in either offense or defense, rather than playing both positions like in other sports. McGee et al. (2012) used two complementary approaches to analyse the efficiency of offensive and defensive players: (i) estimating the Constant Elasticity of Substitution (CES) production function to identify the degree of suitability between offensive strength and the optimal salary allocation; and (ii) estimating the optimal spending ratio directly from the data through a regression model. Offensive and defensive salaries were used as inputs to determine the optimum spending ratios in the NFL. The output was the number of wins in each 16-game regular season between 2000 and 2009. The value of players was assumed to reflect their overall in-game effectiveness. While the old adage says "defense wins championships", the shape of the regression line indicated that the key was to have a clear strategy leaning towards either offense or defense, as balanced spending strategies tended to yield poorer results. Injuries to one or two high-paid superstars could impact the talent ratio of any team.

Roach (2018) examined the effect of player absences on NFL teams by analyzing the percentage of team payroll that was inactive throughout a season. The variable of player injuries was found to significantly predict team performance in terms of wins and point differential. Injuries within individual position groups were analysed to assess the marginal impact of salary resources across specific positions. While one might expect the forces of supply and demand to bring wages to the level of the players' expected marginal revenue, player absences highlighted which positions were undervalued. The salary out rate measured the percentage of a team's payroll, and therefore player talent, that was not utilised throughout a season. Teams with less than 5% of their payroll unavailable tended to perform better, with seven out of 10 teams managing to win 10 out of 16 games over the season. A strong and significant correlation was found between salary out rate and both wins and point differential. However, defensive absences were found

to have less of a detrimental impact on performance. Overall, a one-unit increase in the salary out rate led to 0.22 fewer wins and 6.9 fewer points. Injured players with relatively high market value also showed a higher decline in performance. For offensive players, a one-unit increase in the salary out rate led to 0.3 fewer wins and 9.4 fewer points, compared to 0.13 wins and 3.9 points when defensive talent was absent. Quarterbacks and offensive linemen had the largest impact on overall team performance during the research period, suggesting that these positions may be relatively underpaid despite their high salaries. Defensively, injuries to defensive ends and linebackers had the most significant impact on performance. Interestingly, while elite defensive backs are considered crucial to good defensive performance, more back-ups can fill in, resulting in a less pronounced drop-off in performance (Roach, 2008).

Lozano and Villa (2019) proposed a specific version of the DEA-method that considers both the offence and the defence modes of NFL teams when evaluating their efficiency during the 2016 regular season and the 11 games played in the post-season, including the SuperBowl. The time spent by a team in offence mode versus defence mode during a game was determined by the number of minutes the team possessed the ball. Inputs and outputs were different for the offence and defence mode. Results showed that teams should have spent more time in offence mode than in defence mode. In terms of the efficiency of the two NFL teams that played the Super Bowl, the Patriots, who defeated the Falcons, were overall more efficient due to their higher defensive efficiency. The Falcons were the most efficient team in the two games leading up to the Super Bowl, but they failed to maintain this high level of efficiency in the Super Bowl.

Palafox-Alcantar and Vargas-Hernandez (2015) conducted a study using DEA to analyse the payroll efficiency of the 32 teams in the NFL during the 2014 season. Inputs included total salaries spent on offense and total salaries spent on defense, while outputs consisted of wins, Points Per Game (PPG), Points Allowed Per Game (PAG), YOP and YDP. Offenses were considered payroll inefficient when another team in the league achieved more victories and higher points per game with lower salary expenditure. The study found that teams qualifying for the playoffs were not always the most payroll efficient. Nine out of the 12 playoff teams achieved efficiency in offensive player payroll, with the Carolina Panthers notably struggling and ranking 25th out of the 32 teams in this category. Only four of the 12 playoff teams reached efficiency in terms of defensive player payroll. Some teams were found to overspend on defensive players which wasted resources and impacted their defensive efficiency. Surprisingly, two of the league's weakest teams in terms of wins, Washington and Tennessee, were efficient in both offensive and defensive spending. A notable limitation of the study was that only data from the 2014 season were utilized.

The present study contributes to the literature by analysing the payroll efficiency of NFL teams over multiple NFL seasons (i.e., 2010-2020), a period that has not been previously studied. Specifically, it build upon the multi-season studies conducted by Einolf (2004) and McGee et al. (2012), which analysed payroll efficiency of NFL teams during the periods 1981-2000 and 2000-2009, respectively. Furthermore, it extends the most recent study by Palafox-Alcantar and Vargas-Hernandez (2015) by examining all NFL seasons from 2010-2020. The premise is that a multi-season analysis will yield more informative results.

2.3. DEA research on other sports

Several DEA-studies have been conducted to evaluate team efficiency in sports other than the NFL. Examples include basketball, soccer, and cycling.

Basketball is an interesting sport as it features a salary cap similar to the NFL and requires team management to prioritize specific positions in order to enhance overall performance of the team. Yang, Lin and Chen (2014) utilized a two-stage DEA framework to measure and evaluate the wage efficiency and on-court efficiency of NBA teams. The findings indicated that NBA teams were more efficient in terms of wages (85.22% - 92.62%) than on-court performance (67.14% - 79.1%). Additionally, the results showed that the Western Conference was more efficient than the Eastern Conference, winning 10 out of the 15 NBA Championships after 1998. Interestingly, big market teams like the New York Knicks and Los Angeles Lakers were more efficient in terms of gate receipts compared to small-market teams, regardless of on-court performance. In a study of Yang, Wei, Liang, Ding and Wang (2019) that evaluated NBA team performance using a DEA approach, teams were divided into homogeneous sub-groups due to some teams lacking star players while others played additional playoff games. Inputs included player salary and number of players, as over-investment in players or wages was deemed inefficient. Players were categorized as All-Star or ordinary players, with the outputs being the number of wins in the regular season and playoffs separately. Results revealed, among other things, that the Utah Jazz needed to enhance the efficiency of ordinary players in the playoffs, while the Toronto Raptors performed well in the playoffs but needed to improve their regular season performance. Consistent with previous studies, teams in the Western Conference were found to be more efficient.

Several studies have utilized the DEA method to analyse team efficiency in soccer. Gonzalez-Gomez, Picazo-Tadeo and Garcia-Rubio (2011) applied DEA to measure and compare the efficiency of Spanish football teams that changed managers mid-season in La Liga, the country's top division, from the 2001/02 to 2008/09 seasons. Inputs included team budget and number of spectators, while outputs consisted of points obtained in La Liga and additional matches won in other competitions, like the Spanish Cup or Champions League. Gonzalez-Gomez et al. (2011) concluded that managerial changes generally improved the performances of La Liga teams. Barros and Leach (2006) used DEA to evaluate the sporting and financial best-practice performance of English Premier League clubs from 1998/99 to 2002/03 using data from Deloitte and the Premier League. Inputs were number of players, wages, net assets and stadium expenditure, while outputs were points, attendance and turnover. Results showed that clubs with a large turnover or from cities with higher population bases tended to have higher efficiency scores than their smaller counterparts. Zambom-Ferraresi et al. (2019) employed DEA and bootstrapped DEA methods to analyse the efficiency of Premier League clubs from 2012/13 to 2014/15. Market value, match performance and wages were the inputs in this study, while the outputs were points, revenue, attendance and fan impact on social media. Manchester United was found to be the most efficient club in this period, independent of on-field performance or wages.

Rogge, Van Reeth and Van Puyenbroeck (2013) used a robust (order-m) DEA approach to evaluate the performance of Tour de France cycling teams from 2007-2011. They distinguished between ranking, sprint and mixed teams as there are multiple ways to consider team success. The two inputs considered were team quality and team experience. The outputs were prize money won, Cycling Quotient (CQ) points collected and prizes won. The results showed that ranking teams performed more efficiently compared to sprint and mixed teams (average

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efficiency of 120.1%, 80.1% and 85.4%, respectively). Additionally, larger teams outperformed smaller teams and provided better value for money.

3. Methodology

The suggested approach builds upon DEA, a popular methodology originally developed by Farrell (1957) and further popularized by Charnes, Cooper and Rhodes (1978) in the Operations Research and Management Science literature. DEA is a non-parametric efficiency measurement technique that estimates the relative efficiency of units using (possibly multiple) inputs to produce (possibly multiple) outputs in environments typically characterized by unreliable information on the input and output prices (weights) and/or no exact knowledge of the production or cost function's 'functional form'. In the OR-MS literature, units are generally referred to as decision making units (DMUs). In this case study, the DMUs refer to the NFL teams in the sample. A key feature of the DEA-method is that the efficiency of DMUs is measured in a data-driven way by comparing the input and output data of each DMU relative to the input and output data of all DMUs in the sample. DEA uses linear programming to estimate the relative efficiency of the DMUs. The two basic models for the DEA method are the CCR-DEA model (Charnes, et al., 1978) which assumes constant returns to scale and the BCC-DEA model (Banker, Charnes & Cooper, 1984) which assumes variable returns to scale. Each of these models can be input or output oriented. The output oriented DEA-model looks at the maximum proportional increase in all outputs for the DMU given its current input levels. DMUs producing the highest level of outputs for the current level of inputs are considered efficient and used as benchmarks for all other DMUs. The input oriented DEA-model looks at the maximum proportional decrease in the input levels of the DMU while maintaining at least its current output levels. DMUs using the lowest levels of inputs for producing the current level of outputs are considered efficient and used as benchmarks for all other DMUs. Given that NFL teams aim to perform as well as possible under salary cap restrictions, both offensively and defensively, to increase their chances of success in qualifying for the playoffs, reaching and eventually winning a Super Bowl, the output orientated DEA model is most suitable for evaluating team payroll efficiency. Formally,

$$\min_{c_i, p_r} \sum_{i=1}^{m} c_i x_{i0} - \sum_{r=1}^{s} p_r y_{r0} + v_0 \qquad (1)$$

$$s.t.$$

$$\sum_{i=1}^{m} c_i x_{ij} - \sum_{r=1}^{s} p_r y_{rj} + v_0 \geq 0 \qquad \forall j = 1, \dots, n$$

$$\sum_{r=1}^{s} p_r y_{rj} = 1$$

$$c_i, p_r \geq 0$$

$$v_0 \text{ free in sign}$$

In this model (1), the subscript '0' refers to the team under evaluation, while the subscript 'j' refers to the team j, with $j = 1,\dots,n$ as there are n teams in the dataset (i.e., $n = 32$ as there are 32 teams per NFL-season). x_{i0} refers to the inputs of the team under evaluation, while x_{ij} relates

to the inputs of the team j. y_{r0} (for team 0) and y_{rj} (for team j) refer to the outputs of the teams. c_i is the DEA-estimated input weight for input i for the evaluated team (with i = 1,…,m, m = number of inputs) and p_r is the DEA-estimated output weight for the desirable output r for the evaluated team (with r = 1,…,s, with s = number of desirable outputs). v_0 refers to the free in sign variable related to the variable returns to scale assumption. The model as in (1) is thus computed n times (n = 32) with each computation model evaluating one NFL team relative to all NFL teams in the respective season. Each NFL team is thus evaluated 10 times, i.e. once per NFL-season, by comparing its input and output data relative to the input and output data of all NFL teams in that season. The model (1) assumes convexity of the production technology, a contemporaneous technology approach and strong disposability of the undesirable outputs. Convexity is a fundamental assumption in traditional DEA models. In the present case study, this assumption implies that a team can manage part of its payroll similarly to one team and part of its payroll differently based on another team's ideas.[1] The contemporaneous technology approach assumes that there is a production possibility set for each NFL season and these sets may vary across seasons (e.g., due to changes in team formation through free agency and player trades, changes in coaches, etc.). The assumption of strong disposability of the undesirable outputs (only present in Model 1 and Model 3, i.e. points allowed per game and yards per defensive play) implies that reducing undesirable outputs does not necessarily require a simultaneous reduction in desirable outputs.

The basic DEA-model, as shown in (1), only allows for desirable outputs, meaning outputs where higher values indicate better performance. Therefore, this basic model focuses on identifying potential improvements by determining how much outputs can be increased. In the NFL, offensive statistics like Points Per Game (PPG) and Yards Per Offensive Play (YOP) align with this basic model. However, in terms of defensive performance, NFL teams excel when statistics such as Points Allowed Per Game (PAG) and Yards Per Defensive Play (YDP) are lower. These types of outputs are considered as undesirable, where lower values indicate better performance. Therefore, an adapted DEA model is necessary to also identify improvements in terms of decreasing these output values.

To allow for this juxtaposition, the directional distance DEA Model is used in this study to accurately assess the payroll efficiency of NFL-teams by incorporating both offensive and defensive team performance statistics. The directional distance version of the DEA-model enables the inclusion of both desirable and undesirable outputs in the efficiency evaluation of DMUs (Chung, Färe & Grosskopf, 1997). In this study, the directional distance DEA-approach enables the examination of payroll efficiency by identifying potential improvements in both the team's offensive and defensive statistics simultaneously. Formally,

$$\beta_0^* = \min_{c_i, p_r, q_u} \sum_{i=1}^{m} c_i x_{i0} - \sum_{r=1}^{s} p_r y_{r0}^{+} + \sum_{u=1}^{l} q_u y_{u0}^{-} + v_0 \qquad (2)$$

$$s.t.$$

$$\sum_{i=1}^{m} c_i x_{ij} - \sum_{r=1}^{s} p_r y_{rj}^{+} + \sum_{u=1}^{l} q_u y_{uj}^{-} + v_0 \geq 0 \qquad \forall j = 1, \dots, n$$

[1] It is important to note that convexity is not always without criticism, and that there are alternative methodologies in the literature that either relax or modify this assumption (e.g., Free Disposal Hull and conditional convexity).

$$\sum_{r=1}^{s} p_r g_r^+ + \sum_{u=1}^{l} q_u g_u^- = 1$$
$$c_i, p_r, q_u \geq 0$$
$$v_0 \text{ free in sign}$$

Similar to Model (1), in the directional distance DEA-model as shown in (2), the subscript '0' refers to the NFL team under evaluation, while the subscript 'j' refers to NFL team j with j = 1,…,n (i.e., n = 32 as there are 32 teams per NFL-season). x_{i0} presents the inputs of the NFL team under evaluation, while x_{ij} presents the inputs of the team j. y_{r0}^+ and y_{u0}^- are the offensive performance statistics (desirable outputs, where a higher score is better) and defensive performance statistics (undesirable outputs, where a lower score is better) for team 0. y_{rj}^+ and y_{uj}^- are the offensive and defensive performance statistics for team j. c_i is the DEA-estimated input weight for input i (i = 1,…, m, with m = number of inputs) for the team 0. p_r is the DEA-estimated output weight for the desirable output r (r = 1,…, s, with s = number of offensive performance statistics) and q_u is the DEA-estimated output weight for the undesirable output u (u = 1,…, l, with l = number of defensive performance statistics). v_0 refers to the free in sign variable related to the variable returns to scale assumption in the model.

The team payroll efficiency measure can be computed by $\theta_0^* = 1/(1+\beta_0^*)$, where β_0^* is the objective value of the directional distance DEA model as in (2). The team payroll efficiency measure θ_0^* varies between 0 and 1 (or 100%). In the interpretation, teams that are payroll efficient are evaluated with an estimated θ_0^* equal to a value of one. They do not require reductions in the undesirable outputs and/or increases in the desirable outputs in order to become payroll efficient. The payroll efficient teams act as their own benchmarks. On the other hand, payroll inefficient team are evaluated with θ_0^*-values lower than one, with lower θ_0^*-values indicating more room for improvements in offensive and defensive performance statistics.

In model (2), the direction vector $g = (0, g_r^+, -g_u^-)$ indicates the direction in which desirable outputs and undesirable outputs should be measured for improvements. This vector can take on various values and formats. Färe et al. (2008) discussed several examples of possible formats. In the context of evaluating NFL team payroll efficiency by seeking simultaneous improvements in offensive and defensive performance statistics, one possible direction vector is $g = (0,1,1)$. This suggests that all teams should face the same weights for the offensive and defensive performance statistics when evaluating payroll efficiency. Another option is to set the direction values equal to the current values of desirable and undesirable outputs for the team being evaluated, i.e., $g = \left(0, y_{r0}^+, -y_{u0}^-\right)$. This approach implies unique paths for improvement for each team. A third option for the direction vector is to use the average values of offensive and defensive statistics observed across all teams, i.e. $g = \left(0, \overline{y_{r0}^+}, -\overline{y_{u0}^-}\right)$. It is this latter option that is utilised as the direction vector in the current analysis.

An important potential downside of the large flexibility in the data-driven weighting inherent in the DEA-model, both in its traditional form as in (1) as well as in its directional distance version as in (2), is that zero or quasi-zero weights can be assigned to inputs and (desirable or undesirable) outputs. This results in the neglect of these inputs and outputs in the efficiency

evaluation. To address this issue and acknowledge the significance of each output in the evaluation of team payroll efficiency, a minimalist approach was taken. This approach involved setting an arbitrary minimum contribution requirement from each output. Specifically, weight restrictions were implemented on the desirable and undesirable output weights to ensure that each output receives a minimal level of importance weight in the measurement and evaluation of NFL team payroll efficiency. In the analysis of team payroll efficiency in the context of overall allocation of salary (see Model 1 in Table 1 below) where there are five outputs, a minimum importance weight of at least 10% was imposed on each desirable and undesirable output in the calculation of the overall performance score (refer to (3a) and (3b)):

$$\frac{p_r y_{r0}^{+}}{\sum_{r=1}^{s} p_r y_{r0}^{+} + \sum_{u=1}^{l} q_u y_{u0}^{-}} \geq 0.10 \qquad (3a)$$

$$\frac{q_u y_{u0}^{-}}{\sum_{r=1}^{s} p_r y_{r0}^{+} + \sum_{u=1}^{l} q_u y_{u0}^{-}} \geq 0.10 \qquad (3b)$$

In the other three DEA-models that examine payroll efficiency in spending on offensive stars, defensive stars, and the starting quarterback (see Models 2, 3 and 4 in Table 1), where there are three outputs, similar weight restrictions were applied. However, the minimum importance weight was set at 20% for each desirable and undesirable output.[2,3]

The model as shown in (2) is nonrobust because it does not effectively account for possible outliers in the sample (see, for example, Daraio and Simar 2007 for a discussion). One team in the sample could have data that are extremely outlying or mismeasured, which would downwardly bias the DEA-estimated payroll efficiency scores of all other teams in the sample. To address this issue, this research utilises the robust order-m version of the directional distance DEA model as in (2) (following Cazals, Florens & Simar, 2002). In essence, the robust version of the model calculates payroll efficiency for each team using Monte Carlo simulations. Specifically, for each team a large number of B computation rounds are conducted (in casu, B = 1,000 rounds) in each of which a subsample of randomly selected teams (in the present analysis, m = 20) is considered.[4] This significantly reduces the impact of a single team on the payroll efficiency evaluations of the other teams. The robust team payroll efficiency score is then determined as the average payroll efficiency score calculated over the rounds (refer to Daraio and Simar 2007 for a detailed discussion on the robust analysis). When interpretating the robust team payroll efficiency scores, it is important to note that team payroll efficiency scores are no longer limited to a range between 0 and 1. Specifically, robust payroll efficiency scores higher than 1 indicate that the evaluated team outperforms the best practice payroll

[2] An anonymous referee correctly pointed out that the inclusion of weight restrictions affects the efficiency scores. Specifically, Sarrico and Dyson (2004) found that the use of virtual proportional weight restrictions like those used in the present study, in DEA-models may alter the efficiency scores.

[3] As a robustness check, the estimations were also performed with lower weight bound values set at 5% in Model 1 and 10% in Models 2, 3 and 4, and once without any weight restrictions. Overall, this resulted in only minor differences in the resulting team payroll efficiency scores and rankings of the NFL teams (refer to the web appendices for detailed results).

[4] Daraio and Simar (2007, p. 78) noted that there is no standard methodology for computing the most appropriate value for m. We followed their practical procedure which involves testing various values of m (in this case, all values between 2 and 32) and examining the associated percentages of superefficient performances (teams with payroll efficiency scores indicateing super efficiency).

efficiency performance established by the average comparison set of m randomly selected teams. If the robust payroll efficiency score of the evaluated team is equal to 1, it means that this team performs equally well as the best practice payroll efficiency performance set by the average comparison group. Robust team payroll efficiency scores lower than 1 suggest that the evaluated team falls short of the best practice payroll efficiency performance as determined by the average comparison set of teams. The lower the robust team payroll efficiency score, the greater the opportunity for improvement in the payroll efficiency of the evaluated team.

4. Data

The inputs and outputs for the current payroll efficiency analysis were selected based on previous DEA-based studies on the NFL. Salary cap information from the website Spotrac was used as an input. Previous studies by Einolf (2004), McGee et al. (2012), Palafox-Alcantar and Vargas-Hernandez (2015), and Roach (2018) have used salary cap information as inputs, with total offensive and defensive salaries or specific salaries broken down by player and position. In the present analysis, the inputs for each model were the percentage spent by each of the 32 NFL teams on different aspects of the team. For Model 1, the percentage spent for each team was collected for: (i) top 5 players; (ii) top 6-10 players; (iii) top 11-30 paid players. The percentage allocated to the top five offensive players, top five defensive players and starting quarterbacks were collected for Model 2, Model 3 and Model 4, respectively.

Performance statistics related to the effective performance of NFL teams were selected as outputs. All outputs were gathered from Pro Football Reference, NFL and ESPN. A significant output is regular season wins, which is a clear indicator of team success. Wins have also been a key output metric in studies by Einolf (2004), McGee et al. (2012), Palafox-Alcantar and Vargas-Hernandez (2015) and Roach (2018). Other outputs include performance statistics specifically related to the offense or defense of NFL teams. Points Per Game (PPG) and Yards Per Offensive Play (YOP) are used to measure offensive performance, while Points Allowed Per Game (PAG) and Yards Per Defensive Play (YDP) are used to measure defensive performance. Palafox-Alcantar and Vargas-Hernandez (2015) utilised all four statistics in their research, while Einolf (2004) used YOP and YDP. It is important to note that higher scores for offensive statistics are "desirable", while high scores for defensive statistics are "undesirable". Lastly, a passing-specific statistic in Yards Per Passing Attempt (YPA) will be utilised when assessing the performance of quarterbacks.

Four team payroll efficiency models are investigated, as shown in Table 1. The primary model analyses the overall payroll efficiency of NFL franchises. Each NFL team has 53 players on its roster during the regular season and operates within the same salary cap constraint each season. The offense and defense of NFL teams both have 11 starters who specialise on one side of the ball. Most teams also have specialists for the kicker, punter, long snapper, punt return and kick return positions, while back-up players can play starring roles on some teams. Therefore, the salaries of the top highest-paid players on each NFL team are used as inputs for Model 1, broken down by: (i) top 5 players; (ii) top 6-10 players; (iii) top 11-30 paid players. The aforementioned performance statistics Wins, PPG, PAG, YOP and YDP were used as the outputs of the teams.

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This model investigates if NFL teams were more payroll efficient when allocating substantial financial resources to their star players or spreading funds to the middle of their roster.

The subsequent three DEA-models aim to ascertain if NFL teams who significantly invest in star players on offense, defense or starting quarterback are more payroll efficient. Model 2 and Model 3 analysed the payroll efficiency performance of NFL teams who allocated a high percentage of their salary cap to offensive or defensive players, respectively. Model 4 investigates if NFL teams are more payroll efficient when spending a large percentage of their cap space on quarterbacks, the most important position in the sport. Three outputs were used in Model 4: (i) wins which relate to overall team success; (ii) PPG which shows overall offensive efficiency; and (iii) YPA which shows how effectively the quarterback performed on each play. Data are available for the seasons between 2010 and 2020. However, given that the 2010 season did not have a salary cap due to a labour dispute between the team owners and NFL Players Association, 2010 data are not used and the ten seasons between 2011 and 2020 are analysed in this study. The payroll efficiency of all 32 teams was analysed in each season.

< Insert Table 1 here >

5. Results

The results of the four different DEA models are presented in Tables 2, 4, 5, and 6. The tables are colour coordinated to indicate which teams won the Super Bowl (dark gray and bold), finished runner up (light gray and bold) and made the playoffs (light gray). Each table also includes the number of times a team was identified as a strong efficiency performer, the number of times a team made it to the play offs, the number of times a team was runner up (losing in the Super Bowl final) and the number of Super Bowl wins. The discussion of the results is structured to begin with an analysis of the overall payroll efficiency of the 32 NFL teams over the ten-year period from 2011-2020. It then delves into team payroll efficiency in spending on offensive star players (Model 2), defense star players (Model 3) and starting quarterbacks (Model 4), respectively.

Table 2 displays that the Denver Broncos (five times) and the New England Patriots, Dallas Cowboys and San Francisco 49ers (four times each) consistently outperformed the robust overall payroll efficiency best practice. These four teams combined for 21 playoff appearances, nine Super Bowl appearances and four Super Bowl victories. Generally, teams with high overall payroll efficiency were more successful in terms of playoff and Super Bowl appearances. This result is also evident in Table 3, with the Kruskal Wallis test results revealing that for all NFL seasons in the period 2011-2020, the median payroll efficiency rating of the teams qualifying for the play offs was significantly higher than the median payroll efficiency rating of the teams not qualifying for the play offs. Conversely, Table 2 also demonstrates that the New York Jets, New York Giants, Jacksonville Jaguars, Chicago Bears, Los Angeles Rams and Detroit Lions were consistently inefficient in overall payroll management during the period 2011-2020. The two worst teams in terms of single season overall payroll efficiency were the 2011 Los Angeles Chargers (0.476) and the 2011 Indianapolis Colts (0.502). Notably, these six teams only

qualified for the playoffs 12 times over the 10 year period. Among these bottom performers in overall payroll efficiency, it is interesting to note that the Indianapolis Colts spent the most on top 1-5 and top 6-10 players and the second most on their starting quarterback in 2011, while only spending the 25th most on the 11-30 highest paid players. The 2011 Los Angeles Rams also ranked in the top five in the league in terms of 1-5, 6-10 and the starting quarterback, while ranking in the middle of the pack in terms of 11-30. In these cases, inefficient spending on high-end talent and relatively low expenditure on the middle of the roster negatively affected the overall team performance. Similar observations can be made for other seasons as well (e.g., Kansas City Chiefs in 2012, Houston Texans in 2013, Chicago Bears in 2014, etc.).

< Insert Table 2 here >

< Insert Table 3 here >

Also apparent from Table 2 is that teams who make the Super Bowl tend to have higher overall payroll efficiency ratings. 12 of the 20 teams that made the Super Bowl in the period 2011-2020 ranked in the top 5 of the overall payroll efficiency ranking in the given season, while six ranked in the top 3. Four Super Bowl winning teams, the 2013 Seahawks (1.42343), the 2016 Patriots (1.08456), 2017 Philadelphia Eagles (1.10009) and 2019 Chiefs (1.03479), outperformed the robustly defined best practice in overall payroll efficiency. These consistently high overall payroll efficiency ratings suggest that teams who utilise their salary cap space efficiently are ultimately successful in terms of titles and Super Bowl appearances. A notable outlier is the 2011 New York Giants who won the Super Bowl in that year but ranked 29th in terms of overall payroll efficiency (with an overall payroll efficiency score of 0.83948). An explanation for this result is the observation that this Giants team lost seven regular season games, including five out of six towards the end of the season, while the team also finished with a negative scoring differential. However, the team hit good form in the last two regular season games before making an impressive post-season run to a title. The Giants' 2011 performance suggests their regular season output was not indicative of the overall talent and success level of the team.

< Insert Table 4 here >

< Insert Figure 1 here >

Table 4 displays the team payroll efficiency in spending on offensive stars for the period 2011-2020. As seen in this table, efficient spending on offensive stars had varying results in terms of consistent success between 2011 and 2020. Out of the top seven teams, two teams only made the playoffs three times each despite consistently high payroll efficiency scores. The most significant outlier is the Buffalo Bills. The Bills surpassed the best practices four times in this period (payroll efficiency scores in spending on offensive stars higher than 1) and only had one season in which the team's payroll efficiency in spending on offensive stars was below 0.89, yet the team only won two total playoff games, both in 2020. Similarly, the Tennessee Titans were consistently efficient in terms of payroll spending offensive stars but only made the

playoffs three times. The other eight members of the top ten enjoyed significantly more success, however. These eight teams combined for 55 playoff appearances, 12 Super Bowl appearances and seven Super Bowl victories.

Similar to in Table 2, the New York Jets, Jacksonville Jaguars and New York Giants consistently perform the least payroll efficient. This is clearly illustrated in the radar charts in Figure 1 for the New York Giants and the New York Jets. The Jets are the worst performer with the team scoring a payroll efficiency in spending on offensive stars only twice above 0.74 and never above 0.832. Interestingly, the Jets spent on average the 6th least on star offensive players from 2016-2020, suggesting an underinvestment in that side of the ball. Of the bottom five teams in the payroll efficiency ranking, only the Jacksonville Jaguars obtained a payroll efficiency score higher than 1 (i.e., 1.06421) once, making them appear as a strong performer (see radar chart in Figure 1). Specifically, in 2017, the Jacksonville Jaguars spent the 3rd least on offensive stars that season and ended their playoff drought by making the AFC Championship Game.

Certain teams were consistently inefficient in terms of payroll spending on offensive star players from 2011-2020. The Giants never finished higher than 22nd in the payroll efficiency ranking for spending on offensive star players in a season. Also, note the payroll efficiency ratings of the Los Angeles Rams, which show low figures in the years from 2011 to 2016 before significantly improving in the latter half of the decade. During this time, they made three playoffs and one Super Bowl appearance within four years.

The payroll efficiency ratings in Table 4 suggest that teams that are more efficient in spending on offensive star players are more likely to qualify for the playoffs and achieve post-season success, such as Super Bowl appearances and victories. This positive relationship between team payroll efficiency in spending on offensive star players and play off qualification is further supported by the Kruskal-Wallis test scores for all NFL seasons (refer to Table 3). The top-5 teams in payroll efficiency for spending on offensive stars totaled 31 playoff appearances, 7 Super Bowl appearances and 4 titles. Among these teams, the Buffalo Bills and the New Orleans Saints were the only ones who did not reach the Super Bowl during this period. Interestingly, the only three teams that were outside the top 10 in payroll efficiency for spending on offensive stars in the year they made the Super Bowl were the 2011 Giants, 2012 Ravens and 2015 Broncos, all of whom won the Super Bowl. Despite spending the highest percentage of salary cap space on offensive stars in 2015 (35.6%), the Broncos only ranked 19th in PPG and 18th in YOP. Their success was largely attributed to an impressive defense that ranked 4th in PAG and 1st in YDP.

In terms of specific payroll efficiency ratings in spending on offensive star players, three of the Super Bowl winners scored higher than 1, suggesting strong performance: the 2014 New England Patriots (1.02171), 2016 New England Patriots (1.07978), and 2017 Philadelphia Eagles (1.06308). The 2014 and 2016 Patriots had an established franchise quarterback in Tom Brady who now has six Super Bowl titles. Despite this, he signed a team-friendly contract and was the 12th and 18th highest paid quarterback in 2014 and 2016, respectively. The Eagles won in 2017 with Nick Foles, who was on a back-up contract and received the 7th lowest starting quarterback salary that season. These results demonstrate that exceptional circumstances, such as star players accepting below-market contracts and back-ups exceeding expectations, can lead to high payroll efficiency.

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< Insert Table 4 here >

Table 5 lists the teams' payroll efficiency ratings for spending on defensive star players during the period 2011-2020. The Patriots were the most consistently payroll efficient team during this time. This result is clearly shown in the radar chart in Figure 1. The Patriots made the playoffs nine times, reached the Super Bowl five times, and won three Super Bowls in this period. Other teams that were consistently efficient teams in payroll spending on defensive star players included the Seattle Seahawks (refer to radar chart in Figure 1) and the Pittsburgh Steelers, known for their strong defense.

The Seahawks made the playoffs eight times, appeared in the Super Bowl twice, winning once in 2013. The Steelers made the playoffs six times but did not reach the Super Bowl. The observation that consistent payroll efficiency in spending on defensive star players leads to more postseason success was also noted across other teams. This positive relationship is confirmed by the Kruskal-Wallis test scores in Table 3 for playoff qualifications. In terms of Super Bowl appearances, the model identified five Super Bowl winning teams and six runner up teams as top performers. The winning teams were the 2013 Seahawks (1.07180), 2014 Patriots (1.00684), 2016 Patriots (1.11641), 2017 Eagles (1.00391), and 2019 Chiefs (1.02619). The runner up teams with payroll efficiency scores higher than 1 include the 2011 Patriots, 2013 Broncos, 2014 Seahawks, 2015 Panthers, 2019 49ers, and 2020 Chiefs. Only six teams that made the Super Bowl during this period finished outside the top 10 in payroll efficiency for spending on defensive star players in their respective years: the 2011 Giants, 2015 Panthers, 2016 Falcons, 2017 Eagles, 2018 Rams, and 2020 Buccaneers.

The worst payroll inefficient teams in spending on defensive star players were the Jacksonville Jaguars, Cleveland Browns, Tampa Bay Buccaneers, Las Vegas Raiders and New York Jets. These five teams also combined for a mere four playoff appearances. The lowest payroll efficiency score of any team was 0.5 for the 2017 Browns, when the team only spent 13.5% of their cap space on defensive stars, the second lowest in the league that season, suggesting an underinvestment. Apart from the year 2020, the Browns obtained payroll efficiency ratings lower than 0.8, indicating that there is considerable room for improvement in spending on star defensive players. Similar remarks hold for the other teams in the bottom-5 of the payroll efficiency ranking for spending on defensive star players. The Tampa Bay Buccaneers only finished with a payroll efficiency rating above 0.8 in 2020, the year in which they won the Super Bowl. The Jacksonville Jaguars only obtained a payroll efficiency rating higher than 0.8 in their successful 2017 season when they reached the AFC Championship Game. Additionally, the Las Vegas Raiders and New York Jets only realized a payroll efficiency rating higher than 0.8 once (2016 Raiders and 2015 Jets).

< Insert Table 5 here >

Finally, Table 6 displays the payroll efficiency ratings for NFL teams in spending on starting quarterbacks from 2011-2020 as computed by Model 4. The most consistently payroll efficient

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teams were the Seattle Seahawks, Dallas Cowboys and New England Patriots. Interestingly, these teams had stability at the quarterback position for several seasons, with the Patriots having Tom Brady for 9 seasons, the Seattle Seahawks with Russell Wilson for 9 seasons, and the Dallas Cowboys with Tony Romo and Dak Prescott each for 5 seasons. Other high scoring teams in terms of payroll efficiency in spending on starting quarterback were the Packers, Saints, Broncos and Chiefs (for the radar chart of the Denver Broncos see Figure 1), who had extended franchise quarterbacks in Rodgers, Brees, Manning and Smith and Mahomes, respectively. On the other end of the scale, the New York Jets only scored higher than 0.8 once (2015 Jets). A possible explanation is that the team had 13 different starting quarterbacks over the ten-year period. Other teams that performed poorly in terms of quarterback spending efficiency were the Los Angeles Rams, Chicago Bears and Jacksonville Jaguars. These teams also experienced instability at the quarterback position from 2011-2020, with 10, 11 and 9 starting quarterbacks, respectively. Teams that did have stability at the quarterback position but consistently ranked at the bottom of the payroll efficiency ranking are the New York Giants and Detroit Lions. The Giants had a highly paid starting quarterback in Eli Manning (8 seasons) and the Lions had Matthew Stafford (10 seasons), both of whom did not perform to the level of their contract. Another observation is that out of the 35 times a team was identified as a top performer in payroll efficiency in spending on starting quarterbacks (efficiency rate higher than 1), a significant number of them paid less than 2.5% of the total salary cap on their starting quarterbacks. These players were mainly high draft choices on their rookie contracts, which are predetermined and relatively lower paid compared to established players. This finding suggests that general managers can potentially achieve more payroll efficiency in spending on starting quarterbacks by drafting college players more astutely and benefitting from the low market value.

Also apparent from Table 6 is that teams with efficient payroll spending on their starting quarterback tended to make the playoffs more often. This positive relationship is confirmed by the Kruskal-Wallis test scores in Table 3. Regarding appearances in the Super Bowl, 9 out of 20 teams that made the Super Bowl between 2011 and 2020 ranked in the top five of the payroll efficiency raking for spending on startting quarterbacks. Three notable outliers are Eli Manning (11.61%) in 2011, Joe Flacco (6.63%) in 2012 and Peyton Manning (11.66%) in 2015. Eli Manning and Joe Flacco played for teams that had moderate success in the regular season, the New York Giants and Baltimore Ravens, who finished 9-7 and 10-6 respectively. The Giants and Ravens then reached peak performance during the playoffs to win the Super Bowl. Peyton Manning experienced his worst statistical season in 2015 with the Denver Broncos, recording 17 interceptions and only nine touchdowns. However, the Broncos were carried by a formidable defense, which ranked third in the payroll efficiency ranking for spending on defensive star players in that NFL season (see Table 5 and radar chart in Figure 1) and ranked first in the NFL in PAG and YDP.

6. Conclusions

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Market efficiency is a crucial topic in sports economics and this research aimed to evaluate salary cap efficiency in the NFL. Building on previous DEA research on payroll efficiency in salary cap expenditure in the NFL, this study specifically analysed the payroll efficiency of NFL teams in relation to expenditure on star players over a ten-year period from 2011 to 2020. Four models were defined to analyse payroll efficiency in relation to: (i) the overall allocation of salary; (ii) spending on star offensive players; (iii) spending on star defensive players; and (iv) spending on starting quarterbacks. Inputs included salary cap figures related to the aforementioned categories, while outputs were performance statistics unique to each model. An order-m Directional Distance DEA model was utilised to account for both desirable outputs (team offensive performance statistics) and undesirable outputs (team defensive performance statistics) in the measurement of team payroll efficiency.

Several key lessons were learned that could be used by general managers and head coaches in the NFL. Firstly, payroll efficient teams were found to be more successful in qualifying for the playoffs, making the Super Bowl and winning titles. Of the models analysed in this paper, the model analysing the team payroll efficiency in spending on top defensive players saw the biggest relationship between payroll efficiency and post-season success. However, all models illustrated a consistently prevalent relationship between payroll efficiency and qualifying for the playoffs. Interestingly, teams that qualified for the Super Bowl were mostly highly payroll efficient in all four models. Notable outliers included the 2011 Giants and 2012 Ravens. Teams that performed most consistently payroll efficient in spending on starting quarterback had established franchise quarterbacks who played at a high level for the team for several seasons (stability in the quarterback position) or highly drafted starting quarterback with cost-effective rookie contracts.

This analysis was robust, as it was based on the order-m directional distance DEA model and involved a substantial ten-year sample and multiple outputs. However, certain limitations were present in the analysis. Some of these limitations are related to the specific version of the DEA-method used in the estimations while others are related to the dataset used. Regarding the limitations of the order-m Directional Distance DEA model, we chose to use a multiplier formulation of this model. While the multiplier formulation is well-suited for imposing weight restrictions to avoid (quasi-) zero and/or unrealistically high weights for the outputs in the estimations of team payroll efficiency scores, a drawback of the multiplier version is that it is less suited for computing efficiency target values for the different outputs for the teams. It would be interesting for a follow-up study to derive and use the envelopment formulation of the order-m Directional Distance DEA model to compute efficiency target values for the different outputs (see, for example, Guevel et al. (2024) and Lavigne et al. (2019) for defining output target values). Another interesting extension to the present paper would be to interpret the weight restrictions as team performance trade-offs between the outputs (using insights from Podinovski, 2015). A second methodological limitation relates to the use of percentage input data in the model. It is well-known that the use of percentage data in DEA-models can be problematic (for a discussion, see, for example, Emrouznejad and Amin (2009) and Pastor et al. (2015)).[5] Ideally, one should use versions of the DEA-model that can handle percentage input data. Therefore, it would be interesting for future research to assess the payroll efficiency of NFL teams using such DEA-models (following, for example, Emrouznejad and Amin (2009) and Papaioannou and Podinovski (2023)). Regarding the limitations of the dataset, a larger

[5] We thank an anonymous referee for pointing out this methodological limitation.

sample could be utilised to compare spending efficiency before the salary cap era in 1994 or before rules began to favour offenses in the early 2000s. Additionaly, the positional analysis was restricted to star players generally on offense or defense, but could be expanded to specific positions. For instance, positions such as wide receiver, pass rusher and cornerback tend to command higher salaries, thus posing a risk for inefficient spending. The performance of niche positions like kicker and punter could also be examined. Lastly, the analysis could be extended to the post-season to provide more insights into how efficiently teams performed during the most crucial stage of the season.

The potential for further research in the NFL is vast, with numerous areas that could be explored. One such area is determining the significance of specific positions in achieving team objectives. For instance, it could be examined whether allocating salary towards pass rushers or defensive backs is more effective in disrupting opposing offenses, or if investing heavily in the offensive line leads to improved performance in passing or rushing plays. Additionally, the impact of special teams, often overlooked, could be assessed by evaluating the success of teams that prioritize special teams players. Another potential area of study is the relation between a team's average squad age and their efficiency in managing the salary cap, as teams strive to strike a balance between youthful talent and veteran experience. Furthermore, further analysis could be conducted to determine if teams become more payroll efficient after replacing general managers or head coaches.

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Compliance with Ethical Standards

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Figure 1: Radar charts with overall, offensive stars, defensive starts and starting quarterback payroll efficiency

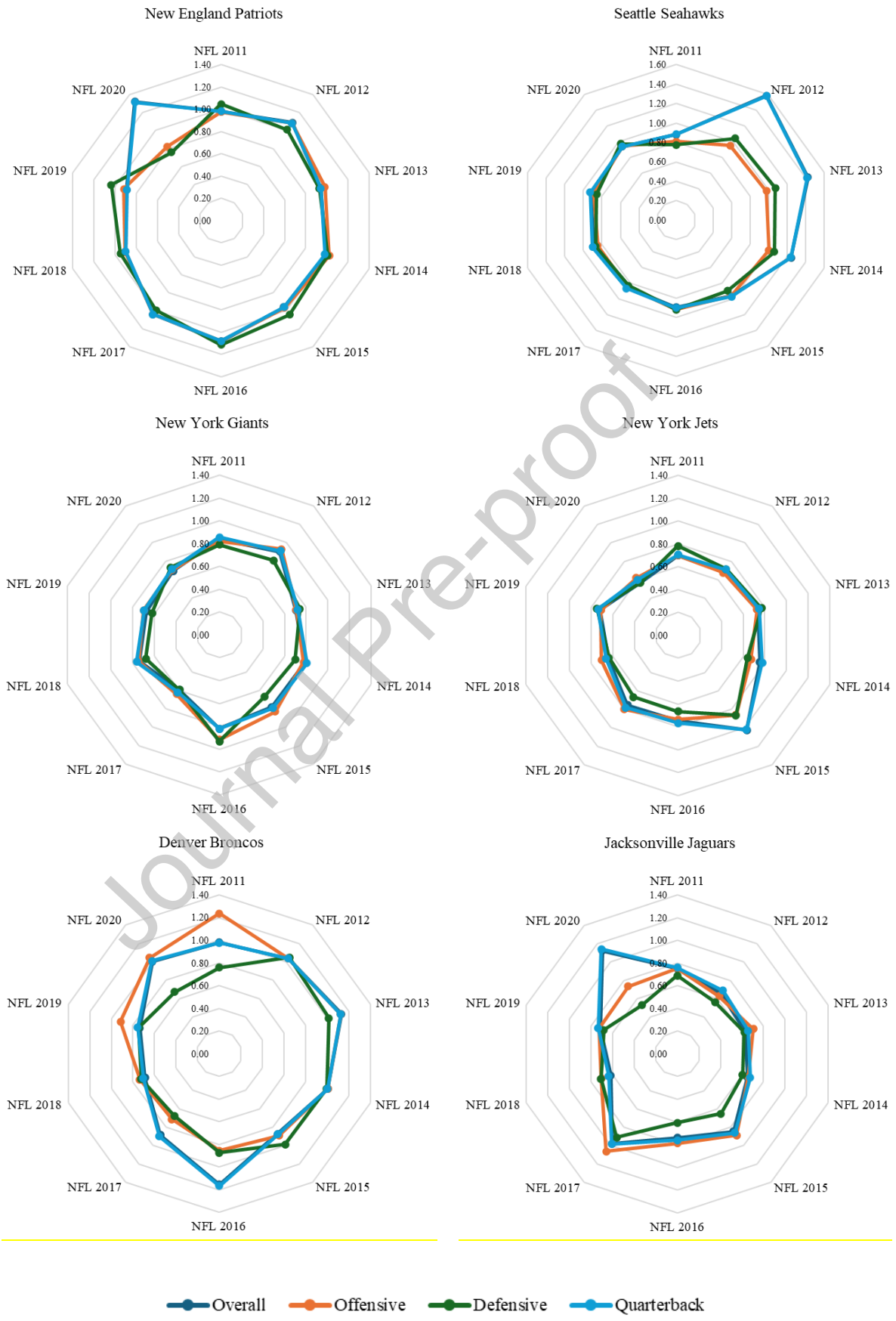

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Table 1: Overview of DEA-Models

Model	Inputs	Outputs
1: Overall Payroll Efficiency	(a) Top 5 Players (b) Top 6-10 Players (c) Top 11-30 Players	(1) Wins (2) PPG (3) PAG (4) YOP (5) YDP
2. Payroll Efficiency Offensive Stars	Top 5 Offensive Players	(1) Wins (2) PPG (3) YOP
3: Payroll Efficiency Defensive Stars	Top 5 Defensive Players	(1) Wins (2) PAG (3) YDP
4. Payroll Efficiency Starting Quarterback	Starting Quarterback	(1) Wins (2) PPG (3) YPA

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Table 2: Overall payroll efficiency

City	Team	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	Strong performer	Play Offs	SB Winner	Runner Up
Arizona	Cardinals	0.86628	0.60092	0.90909	0.8711	1.07885	0.82553	0.7597	0.60593	0.75354	0.82696	1	2	0	0
Atlanta	Falcons	0.79464	1.00962	0.70023	0.79269	0.79128	**1.07269**	0.90451	0.82636	0.79832	0.74161	2	4	0	1
Baltimore	Ravens	0.93235	**0.88641**	0.73713	0.87214	0.67536	0.73021	0.80832	0.80041	1.24826	1.14878	2	6	1	0
Buffalo	Bills	0.78725	0.7596	0.75672	0.86891	1.33166	0.89238	0.79826	0.68599	0.83744	1.05898	2	3	0	0
Carolina	Panthers	0.92085	0.88998	0.9371	0.81414	**1.07698**	0.75037	0.86089	0.7544	0.67591	0.75613	1	4	0	1
Chicago	Bears	0.74763	0.84958	0.89337	0.68401	0.74397	0.6819	0.67862	0.96143	0.71719	0.76013	0	2	0	0
Cincinnati	Bengals	1.3255	1.12131	1.04278	0.89776	0.97033	0.80685	0.73525	0.70185	0.59491	0.67509	3	5	0	0
Cleveland	Browns	1.14529	0.82523	0.69658	0.89663	0.72847	0.67771	0.78071	0.77862	0.76508	0.91139	1	1	0	0
Dallas	Cowboys	0.86578	0.86037	0.8549	1.02843	0.65812	0.99764	1.38341	1.74008	1.2144	0.757	4	3	0	0
Denver	Broncos	0.97873	1.03872	**1.13258**	1.00522	**0.87619**	1.15413	0.88502	0.68851	0.73661	1.00352	5	5	1	1
Detroit	Lions	0.95623	0.72946	0.79356	0.87021	0.76206	0.81276	0.91265	0.6805	0.70115	0.76564	0	3	0	0
Green Bay	Packers	1.13892	0.96731	0.88763	1.02067	0.80075	0.90543	0.72445	0.74797	0.9251	1.02349	3	8	0	0
Houston	Texans	0.84152	0.96041	0.60096	0.91258	0.83688	0.74359	0.90123	1.07273	0.95259	0.85567	1	6	0	0
Indianapolis	Colts	0.50185	0.96212	0.90732	1.02216	0.75451	0.86086	0.65701	0.83801	0.75261	0.91406	1	5	0	0
Jacksonville	Jaguars	0.75231	0.65712	0.63416	0.6485	0.84468	0.7395	0.98008	0.61638	0.71432	1.12323	1	1	0	0
Kansas City	Chiefs	0.68612	0.56818	0.90669	0.8745	0.88279	0.92983	0.9487	1.17642	**1.03479**	**1.08809**	3	7	1	1
Las Vegas	Raiders	1.02639	0.7353	0.8649	0.56621	1.07299	1.4436	0.72594	0.6661	0.79737	0.87566	3	1	0	0
Los Angeles	Chargers	0.79064	0.76317	0.88477	0.84106	0.69654	0.79824	0.88577	0.95701	0.75998	0.8284	2	1	0	0
Los Angeles	Rams	0.47582	0.75043	0.72999	0.74768	0.76267	0.69251	1.04794	**1.0657**	0.85594	0.8269	1	4	0	1
Miami	Dolphins	0.88295	0.83077	0.79485	0.897	0.81192	0.94066	0.68598	0.74814	0.70176	0.88153	0	1	0	0
Minnesota	Vikings	0.85992	0.96279	0.79091	0.87409	1.18414	0.86894	0.98433	0.77004	0.90316	0.87082	1	4	0	0
New England	Patriots	**0.97926**	1.0871	0.94489	**0.98479**	0.95911	**1.08456**	**1.03922**	**0.90006**	0.89099	1.31905	4	9	3	2
New Orleans	Saints	1.00068	0.90051	0.91588	0.82272	0.82701	0.90496	0.98987	1.01057	0.9787	0.95417	2	6	0	0
New York	Giants	**0.83948**	0.90193	0.70426	0.78638	0.781	0.82135	0.59448	0.73884	0.67216	0.69122	0	2	1	0
New York	Jets	0.69794	0.69533	0.7335	0.75403	1.02398	0.7455	0.75437	0.64358	0.7201	0.56767	1	0	0	0
Philadelphia	Eagles	0.78469	0.65887	0.97683	1.28931	0.76869	0.87481	**1.10009**	0.82376	0.81962	0.64865	2	4	1	0
Pittsburgh	Steelers	0.85315	0.79904	0.80907	0.9652	0.91934	0.89417	1.00181	0.84511	0.70048	0.88037	1	6	0	0
San Francisco	49ers	1.05791	**0.97585**	1.05239	0.83947	0.65549	0.61723	1.32035	0.73221	**1.03038**	0.78176	4	4	0	2
Seattle	Seahawks	0.87791	1.5803	**1.42253**	**1.24225**	0.96125	0.89536	0.85512	0.89005	0.91619	0.93506	3	8	1	1
Tampa Bay	Buccaneers	0.8247	0.8275	0.93103	0.67225	0.85756	0.94748	0.82132	0.83066	0.876	**0.94907**	0	1	1	0
Tennessee	Titans	0.8516	0.73405	0.80711	0.70729	0.74711	1.00229	0.86788	0.80209	0.93661	0.95691	1	3	0	0
Washington	Football Team	1.01698	1.02649	0.67043	0.81533	0.89237	0.89253	0.82493	0.67333	0.61823	0.70007	2	3	0	0

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Table 3: Kruskal-Wallis tests for payroll efficiency ratings of non-playoff teams versus playoff teams

NFL season	Overall	Offensive Star Players	Defensive Star Players	Starting Quarterbacks
2011	0.0142	< 0.001	< 0.001	0.0176
2012	< 0.001	< 0.001	< 0.001	< 0.001
2013	< 0.001	< 0.001	< 0.001	< 0.001
2014	0.0014	< 0.001	< 0.001	0.0024
2015	0.0011	0.0064	< 0.001	0.0016
2016	0.0114	0.0040	< 0.001	0.0265
2017	0.0021	< 0.001	< 0.001	0.0027
2018	< 0.001	< 0.001	< 0.001	< 0.001
2019	< 0.001	< 0.001	< 0.001	< 0.001
2020	0.0167	0.0021	< 0.001	0.0276

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Table 4: Payroll efficiency offensive stars

City	Team	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	Strong performer	Play Offs	SB Winner	Runner Up
Arizona	Cardinals	0.70052	0.58961	0.93149	0.89455	1.07462	0.83662	0.72671	0.57242	0.75541	0.86523	1	2	0	0
Atlanta	Falcons	0.8072	0.99975	0.7132	0.81026	0.8059	**1.06585**	0.90508	0.82882	0.78551	0.72158	2	4	0	1
Baltimore	Ravens	0.96477	**0.92471**	0.74859	0.92749	0.71739	0.81959	0.81862	0.81409	1.21587	1.1092	2	6	1	0
Buffalo	Bills	1.08522	0.95867	0.92471	1.02816	1.0721	0.93762	0.76431	0.99681	0.8903	1.06333	2	3	0	0
Carolina	Panthers	0.78731	0.8303	1.01987	0.80112	**1.07405**	0.7403	0.86771	0.79335	0.70124	0.76116	1	4	0	1
Chicago	Bears	0.86334	0.83671	0.9318	0.70838	0.74406	0.71402	0.70198	0.98112	0.70724	0.79275	0	2	0	0
Cincinnati	Bengals	0.80005	0.98126	1.11212	0.93286	0.93832	0.76761	0.72584	0.72141	0.58948	0.69064	3	5	0	0
Cleveland	Browns	0.53002	0.73158	0.82471	0.80871	0.7604	0.87981	0.63472	0.80483	0.77816	0.92697	1	1	0	0
Dallas	Cowboys	0.95815	0.91165	0.9103	1.01549	0.71881	1.00672	0.83183	0.83029	0.91265	0.74528	4	3	0	0
Denver	Broncos	1.23423	1.03563	**1.12638**	1.00685	**0.89468**	0.85573	0.71332	0.73826	0.91557	1.04617	5	5	1	1
Detroit	Lions	0.87088	0.75977	0.81559	0.86594	0.77553	0.83076	0.88115	0.68401	0.68841	0.78875	0	3	0	0
Green Bay	Packers	1.08005	0.9715	0.92368	1.02354	0.82179	0.92029	0.73735	0.76868	0.92366	1.01979	3	8	0	0
Houston	Texans	0.81836	0.9485	0.60797	0.81435	0.873	0.78197	0.71798	1.01809	0.90885	0.82588	1	6	0	0
Indianapolis	Colts	0.53057	1.37925	0.96324	1.1263	0.92341	0.83817	0.62171	0.86343	0.77816	0.9146	1	5	0	0
Jacksonville	Jaguars	0.75423	0.62911	0.70573	0.66081	0.88853	0.78823	1.06421	0.70516	0.72792	0.73461	1	1	0	0
Kansas City	Chiefs	0.71824	0.60005	0.96778	0.82404	0.90547	0.9435	0.9783	1.06489	**0.9768**	**1.03792**	3	7	1	1
Las Vegas	Raiders	0.7853	0.78425	1.02808	0.65404	0.86869	0.98855	0.75216	0.65065	0.78566	0.849	3	1	0	0
Los Angeles	Chargers	0.81037	0.7386	0.88115	0.82355	0.70041	0.80478	0.89325	0.95145	0.75625	0.82538	2	1	0	0
Los Angeles	Rams	0.48499	0.77167	0.78307	0.74236	0.81333	0.68511	1.00088	**1.08397**	0.84514	0.84206	1	4	0	1
Miami	Dolphins	0.66473	0.74516	1.04618	0.84442	0.8819	0.92794	0.70036	0.74706	0.94321	1.072	0	1	0	0
Minnesota	Vikings	0.62033	0.92218	0.80414	0.74621	0.85589	0.76587	0.99619	0.78088	0.87748	0.86699	1	4	0	0
New England	Patriots	**0.97574**	1.08174	0.97834	**1.02171**	0.96639	**1.07978**	**1.03723**	**0.90692**	0.91901	0.8215	4	9	3	2
New Orleans	Saints	1.05747	0.96871	0.92231	0.85685	0.84213	1.05738	1.00693	1.00874	0.97513	0.98051	2	6	0	0
New York	Giants	**0.81846**	0.92853	0.70741	0.77571	0.83198	0.91692	0.63906	0.74588	0.69781	0.70337	0	2	1	0
New York	Jets	0.69769	0.67605	0.72678	0.67187	0.86291	0.73754	0.79927	0.70321	0.71016	0.61959	1	0	0	0
Philadelphia	Eagles	0.82343	0.69069	0.95642	0.99435	0.83682	0.88188	**1.06308**	0.81389	0.82665	0.67065	2	4	1	0
Pittsburgh	Steelers	0.90376	0.87735	0.84485	0.97561	0.96285	0.91776	0.9993	0.87438	0.69487	0.92302	1	6	0	0
San Francisco	49ers	1.01153	**0.982**	0.97543	0.80835	0.66542	0.68656	1.06756	0.70767	**1.01257**	0.78789	4	4	0	2
Seattle	Seahawks	0.81012	0.94639	**0.97588**	**1.00022**	0.96107	0.91453	0.85063	0.84562	0.89114	0.93847	3	8	1	1
Tampa Bay	Buccaneers	0.6473	0.84576	0.68279	0.63719	0.89886	0.86094	0.8611	0.79503	0.85227	**0.95801**	0	1	1	0
Tennessee	Titans	0.77888	0.74243	0.78237	0.6989	1.10598	1.15775	1.0663	0.91827	0.89284	0.96779	1	3	0	0
Washington	Football Team	0.73715	0.97485	0.69557	0.73222	0.85198	0.90925	0.78335	0.67418	0.61126	0.72237	2	3	0	0

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Table 5: Payroll efficiency defensive stars

City	Team	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	Strong performer	Play Offs	SB Winner	Runner Up
Arizona	Cardinals	0.79004	0.68065	0.86669	0.93091	0.97093	0.78777	0.78786	0.60895	0.66079	0.80169	1	2	0	0
Atlanta	Falcons	0.84153	1.025	0.61358	0.69516	0.83128	**0.98119**	0.87593	0.73881	0.71702	0.63068	2	4	0	1
Baltimore	Ravens	0.99293	**0.8647**	0.77377	0.90023	0.66019	0.80227	0.85903	0.94903	1.03946	0.96525	2	6	1	0
Buffalo	Bills	0.6878	0.69335	0.70602	0.88056	0.75929	0.73941	0.83155	0.74292	0.91312	1.01758	2	3	0	0
Carolina	Panthers	0.6733	0.74847	1.02005	0.77124	**1.07692**	0.7087	0.887	0.74413	0.64668	0.67673	1	4	0	1
Chicago	Bears	0.77833	0.92078	0.7198	0.65889	0.68846	0.60178	0.68178	1.04495	0.79841	0.79805	0	2	0	0
Cincinnati	Bengals	0.84559	0.91644	0.92034	0.89518	0.96114	0.74041	0.74929	0.6959	0.55879	0.64764	3	5	0	0
Cleveland	Browns	0.64657	0.67645	0.63784	0.76816	0.58651	0.53105	0.5	0.80121	0.68918	0.91476	1	1	0	0
Dallas	Cowboys	0.77473	0.77402	0.72984	1.01449	0.62753	0.99957	0.84945	0.89832	0.78745	0.70154	4	3	0	0
Denver	Broncos	0.76039	1.05002	**1.01424**	0.99673	**0.98747**	0.87213	0.67668	0.72422	0.74558	0.67064	5	5	1	1
Detroit	Lions	0.83609	0.63109	0.72428	0.97121	0.71829	0.80703	0.79649	0.72142	0.60489	0.65	0	3	0	0
Green Bay	Packers	1.0362	0.91956	0.75116	0.9693	0.84449	0.82689	0.72824	0.73507	1.04411	1.00048	3	8	0	0
Houston	Texans	0.89158	1.01267	0.56407	0.8594	0.82708	0.83802	0.62189	0.94528	0.81122	0.62599	1	6	0	0
Indianapolis	Colts	0.5618	0.88187	0.86809	0.92476	0.74416	0.76736	0.62495	1.01613	0.72927	0.94285	1	5	0	0
Jacksonville	Jaguars	0.68878	0.56476	0.61961	0.60332	0.65356	0.60893	0.91008	0.7085	0.68168	0.53051	1	1	0	0
Kansas City	Chiefs	0.74128	0.56279	0.88066	0.87438	0.93297	0.94321	0.83622	0.9336	**1.02619**	**1.03322**	3	7	1	1
Las Vegas	Raiders	0.74403	0.6274	0.63594	0.59988	0.72277	0.8908	0.69526	0.62729	0.73302	0.77675	3	1	0	0
Los Angeles	Chargers	0.77374	0.77487	0.80379	0.88337	0.61909	0.66953	0.84395	0.99452	0.6711	0.74549	2	1	0	0
Los Angeles	Rams	0.56312	0.76878	0.73045	0.71693	0.74098	0.64095	0.88627	**0.99102**	0.81389	0.95412	1	4	0	1
Miami	Dolphins	0.71264	0.76416	0.77865	0.79611	0.68898	0.84044	0.69611	0.73261	0.64115	0.87662	0	1	0	0
Minnesota	Vikings	0.59134	0.86816	0.66202	0.76191	0.94694	0.80955	1.0494	0.85017	0.86698	0.72099	1	4	0	0
New England	Patriots	**1.045**	1.00774	0.92814	**1.00684**	1.04555	**1.11641**	**0.99375**	**0.95085**	1.03598	0.75807	4	9	3	2
New Orleans	Saints	0.96326	0.70897	0.9038	0.78214	0.72469	0.72144	0.9502	1.06937	1.01774	1.0775	2	6	0	0
New York	Giants	**0.79239**	0.80508	0.74192	0.89823	0.6709	0.93388	0.59529	0.67724	0.6207	0.72906	0	2	1	0
New York	Jets	0.78273	0.71614	0.76814	0.6419	0.86472	0.66719	0.66577	0.63772	0.74753	0.56682	1	0	0	0
Philadelphia	Eagles	0.78082	0.62986	0.92364	0.90327	0.71456	0.74853	**1.00391**	0.83754	0.81914	0.65939	2	4	1	0
Pittsburgh	Steelers	1.02764	0.82806	0.76827	0.8923	0.87512	0.89969	1.07954	0.88216	0.81237	1.00835	1	6	0	0
San Francisco	49ers	1.06886	**0.98937**	0.98358	0.81007	0.65543	0.56094	0.74054	0.64447	**1.01498**	0.73089	4	4	0	2
Seattle	Seahawks	0.77318	1.03749	**1.0718**	**1.05912**	0.8958	0.91879	0.83144	0.86933	0.85824	0.968	3	8	1	1
Tampa Bay	Buccaneers	0.61341	0.73741	0.62742	0.56894	0.69339	0.80032	0.65423	0.66308	0.72792	**0.93259**	0	1	1	0
Tennessee	Titans	0.83362	0.70213	0.73305	0.56805	0.59266	0.81302	0.82017	0.87574	0.8135	0.89149	1	3	0	0
Washington	Football Team	0.66451	0.87592	0.58692	0.6291	0.82607	0.79058	0.73307	0.75948	0.59149	0.79	2	3	0	0

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Table 6: Payroll efficiency starting quarterbacks

City	Team	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	Strong performer	Play Offs	SB Winner	Runner Up
Arizona	Cardinals	0.87825	0.61763	0.91357	0.87105	1.078	0.82594	0.77287	0.62885	0.77198	0.83884	1	2	0	0
Atlanta	Falcons	0.80051	1.00899	0.71879	0.81079	0.80845	**1.07415**	0.91686	0.84367	0.81558	0.76817	2	4	0	1
Baltimore	Ravens	0.92924	**0.89153**	0.7428	0.87362	0.69527	0.7343	0.80462	0.80171	1.23932	1.14239	2	6	1	0
Buffalo	Bills	0.79549	0.77425	0.76434	0.87256	1.33445	0.89836	0.80709	0.69929	0.84642	1.05775	2	3	0	0
Carolina	Panthers	0.93719	0.91024	0.93515	0.82617	**1.07233**	0.76434	0.8608	0.76813	0.69446	0.78387	1	4	0	1
Chicago	Bears	0.75968	0.85436	0.90113	0.70182	0.76622	0.71482	0.702	0.95828	0.72833	0.76853	0	2	0	0
Cincinnati	Bengals	1.31361	1.11811	1.03806	0.89832	0.97394	0.82797	0.75139	0.71348	0.62539	0.69862	3	5	0	0
Cleveland	Browns	1.15614	0.84181	0.71531	0.91356	0.75537	0.71528	0.81914	0.79323	0.78621	0.91734	1	1	0	0
Dallas	Cowboys	0.88111	0.87701	0.86076	1.02943	0.68729	0.99764	1.36798	1.74068	1.22406	0.77543	4	3	0	0
Denver	Broncos	0.97825	1.03759	**1.12666**	1.00356	**0.87864**	1.16588	0.89953	0.70397	0.75478	1.00852	5	5	1	1
Detroit	Lions	0.95557	0.75053	0.80507	0.87056	0.77817	0.82005	0.92481	0.69666	0.73049	0.79215	0	3	0	0
Green Bay	Packers	1.13779	0.97356	0.89839	1.02089	0.80509	0.90446	0.73537	0.76274	0.92392	1.02371	3	8	0	0
Houston	Texans	0.85268	0.9597	0.63079	0.91709	0.83931	0.74704	0.91985	1.0758	0.96408	0.89638	1	6	0	0
Indianapolis	Colts	0.52819	0.95944	0.90499	1.02095	0.7587	0.8722	0.68388	0.83948	0.76393	0.91884	1	5	0	0
Jacksonville	Jaguars	0.75928	0.68728	0.65753	0.67432	0.86088	0.76289	0.98273	0.63758	0.73455	1.14083	1	1	0	0
Kansas City	Chiefs	0.7052	0.60146	0.90192	0.88033	0.88774	0.92914	0.95924	1.17218	**1.04078**	1.08533	3	7	1	1
Las Vegas	Raiders	1.03424	0.75928	0.88474	0.58574	1.07332	1.43832	0.74532	0.69502	0.82322	0.89351	3	1	0	0
Los Angeles	Chargers	0.80604	0.77869	0.90021	0.85209	0.72571	0.81783	0.90164	0.96217	0.78959	0.84662	2	1	0	0
Los Angeles	Rams	0.50429	0.76779	0.73926	0.767	0.77079	0.71899	1.04968	**1.06274**	0.8671	0.83507	1	4	0	1
Miami	Dolphins	0.89822	0.84472	0.80349	0.89957	0.82942	0.95072	0.70273	0.76602	0.72475	0.88706	0	1	0	0
Minnesota	Vikings	0.87077	0.95632	0.79471	0.88627	1.17849	0.88088	0.98438	0.78013	0.9155	0.89424	1	4	0	0
New England	Patriots	**0.98088**	1.08188	0.94069	**0.97877**	0.96118	**1.08241**	**1.03882**	**0.90286**	0.89031	1.31165	4	9	3	2
New Orleans	Saints	0.99846	0.91172	0.92302	0.83631	0.84532	0.90811	0.99532	1.00851	0.97969	0.95377	2	6	0	0
New York	Giants	**0.8562**	0.90883	0.72093	0.80358	0.79368	0.82319	0.62096	0.76218	0.69509	0.7124	0	2	1	0
New York	Jets	0.7044	0.71384	0.74676	0.77566	1.021	0.76791	0.78011	0.66217	0.73783	0.60016	1	0	0	0
Philadelphia	Eagles	0.79889	0.68593	0.98765	1.28449	0.78163	0.87733	**1.09426**	0.83533	0.82463	0.66755	2	4	1	0
Pittsburgh	Steelers	0.86271	0.81323	0.81874	0.97022	0.93202	0.89471	1.00185	0.85118	0.71196	0.87618	1	6	0	0
San Francisco	49ers	1.05326	**0.98512**	1.04922	0.85157	0.68361	0.64355	1.3218	0.76344	**1.03321**	0.80551	4	4	0	2
Seattle	Seahawks	0.88471	1.57986	**1.4149**	**1.23814**	0.96944	0.90286	0.86519	0.89936	0.92542	0.9354	3	8	1	1
Tampa Bay	Buccaneers	0.84153	0.84326	0.94811	0.70473	0.87605	0.95578	0.85097	0.85942	0.8933	**0.9489**	0	1	1	0
Tennessee	Titans	0.85976	0.75023	0.81629	0.7457	0.77478	1.01024	0.8781	0.81499	0.95808	0.95943	1	3	0	0
Washington	Football Team	1.03277	1.03146	0.6925	0.85172	0.9007	0.90673	0.84559	0.68749	0.64846	0.71248	2	3	0	0

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